METALS 13

Metals are found on the left-hand side and centre of the Periodic Table. Most metals are lustrous (shiny) and they are often silver in colour. Metals are good thermal and electrical conductors, whereas most non-metals are poor conductors.

1 What is meant by *a good thermal conductor*?

Metals are malleable; they can be beaten into shape. They are also ductile; they can be drawn into wires. Non-metals are brittle; they snap instead of bend. Metals are usually denser than non-metals. Metals are solids at room temperature (with the exception of mercury, Hg). They usually have high melting points. Very few metals occur **native** (uncombined) in the natural state. They are usually found combined with other elements. The earliest metals to be used by humans were copper (Cu), tin (Sn), silver (Ag) and gold (Au) – all of these metals can be found native.

2 Explain why copper, tin, silver and gold can be found native.

Very few metals occur uncombined in the natural state. They are usually found chemically joined with other elements in **minerals**. A mineral is a naturally-occurring pure substance. An ore is a material from the Earth's crust that contains enough of a substance to make it worth mining. Iron (Fe) exists in the forms of haematite and magnetite, which are both forms of iron oxide (i.e. iron combined with oxygen in a compound). Iron ore is very similar in chemical composition to iron rust.

Aluminium (Al) is found as bauxite (aluminium oxide) and lead (Pb) as galena (lead sulfide, lead combined with sulfur). Copper occurs in the ores malachite and azurite, which are both forms of copper carbonate. Another important ore of copper is chalcopyrite, a yellow mineral consisting of sulfides of copper and iron.

3 Define each term.

a lustrous
b malleable
c ductile
d brittle
e ore
f mineral

4 Give the chemical name for each substance.

a haematite
b bauxite
c galena
d limestone
e malachite
f quicklime

5 Give the names and symbols of the elements present in each substance.

a iron oxide
b aluminium oxide
c lead sulphide
d copper carbonate
e hydrogen peroxide
f ammonia

6 Explain why ores are non-renewable resources.

Mining

Lead was the first metal mined in Australia, in Glen Osmond (South Australia) in 1841. The mining and minerals industry in Australia is very important to the economy. It employs about 200 000 people and is indirectly responsible for another 300 000 jobs. It creates employment and brings in valuable foreign currency. However, despite the economic advantages that mining brings, the mining and minerals industry does cause problems.

7 Define *mining*.

8 Outline some possible negative aspects to mining.

9 Name three substances that are mined in Australia.

10 Outline two benefits of mining.

11 Often *raw* materials are mined. What does *raw* mean in this context?

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